Additive Build Advisor — Technical Report

Seymur Hasanov

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1 Technical Report: Additive Build Advisor

1.1 Executive summary

The Additive Build Advisor is a compact, design-to-inspection digital thread for additive manufacturing. It ingests a part as an STL mesh, recovers a clean geometry, chooses a build orientation by resting the part on candidate flat faces, simulates the build on a voxel model, solves a finite-element **distortion analysis** (the inherent-strain method), checks manufacturability, turns the part's tolerances into an inspection plan, and assembles a single machine-readable record with an explicit release gate.

The goal is not to be a production build processor. It is to demonstrate the engineering workflow behind one, end to end and from first principles:

1. Recover geometry you can trust (parse, re-normal, verify watertightness).

2. Choose orientation physically (which flat face goes down), scored on real support volume, base contact, and height.
3. Simulate the build on a model you can *validate* against analytic volume.
4. Predict distortion with a real **finite-element** solve, validated against an analytical case.
5. Decide manufacturability with transparent DfAM rules.
6. Connect design tolerances to a concrete, capability-aware inspection plan.
7. Gate the physical action — release, review, or redesign — with the reasons.

1.2 Why this project

It fills a specific gap: a runnable artifact that shows the **front half of the digital thread** — design intent turning into a build decision — grounded in real additive-manufacturing physics (orientation, support, FEA distortion, DfAM, process capability) rather than a black-box model. It is built to hand off to a separate runtime-monitoring twin (`mini-manufacturing-digital-twin`), so the two together span design → build → monitor.

The STL parser, geometry kernel, voxelizer, orientation search, and build simulation are written from scratch on `numpy` so that engineering is legible. The distortion FEA is assembled and solved with **scikit-fem** (an established finite-element library) on top of `scipy` — using a real FEM library here is the honest, credible choice; `matplotlib` renders the report figures.

1.3 System architecture

1.4 The voxelization engine

A triangle soup is not a solid model, so the advisor builds an occupancy grid by *ray stabbing*: for each (x, y) column it shoots a vertical ray, collects where it crosses the mesh, sorts the crossings, and fills voxels between entry/exit pairs (the even-odd rule). Per-axis grid jitter prevents sample points from landing on face diagonals; coincident-crossing de-duplication keeps the pairing correct on shared edges; and z is sampled at voxel centers so the discretized volume is unbiased. The grid drives support, thin-wall, trapped-void, and per-layer cross-section estimates, and supplies the element mesh for the FEA.

1.4.1 Engine validation

The voxel/mesh volume error is reported in every run and feeds the gate's simulation confidence.

Axis-aligned cube (10 mm, analytic 1000 mm³) — lands exactly on the grid:

grid_n	voxel volume	error
16	1000.00	0.00%
32	1000.00	0.00%
64	1000.00	0.00%

Off-axis rotated bracket (analytic 8640 mm³) — converges as it refines, staying under ~0.4% even when coarse (0.18% → 0.03% → −0.02% for grid_n 24 → 48 → 96). A solid cube reports zero trapped/zero support; a hollow box with a 216 mm³ void recovers ~220 mm³ by flood fill.

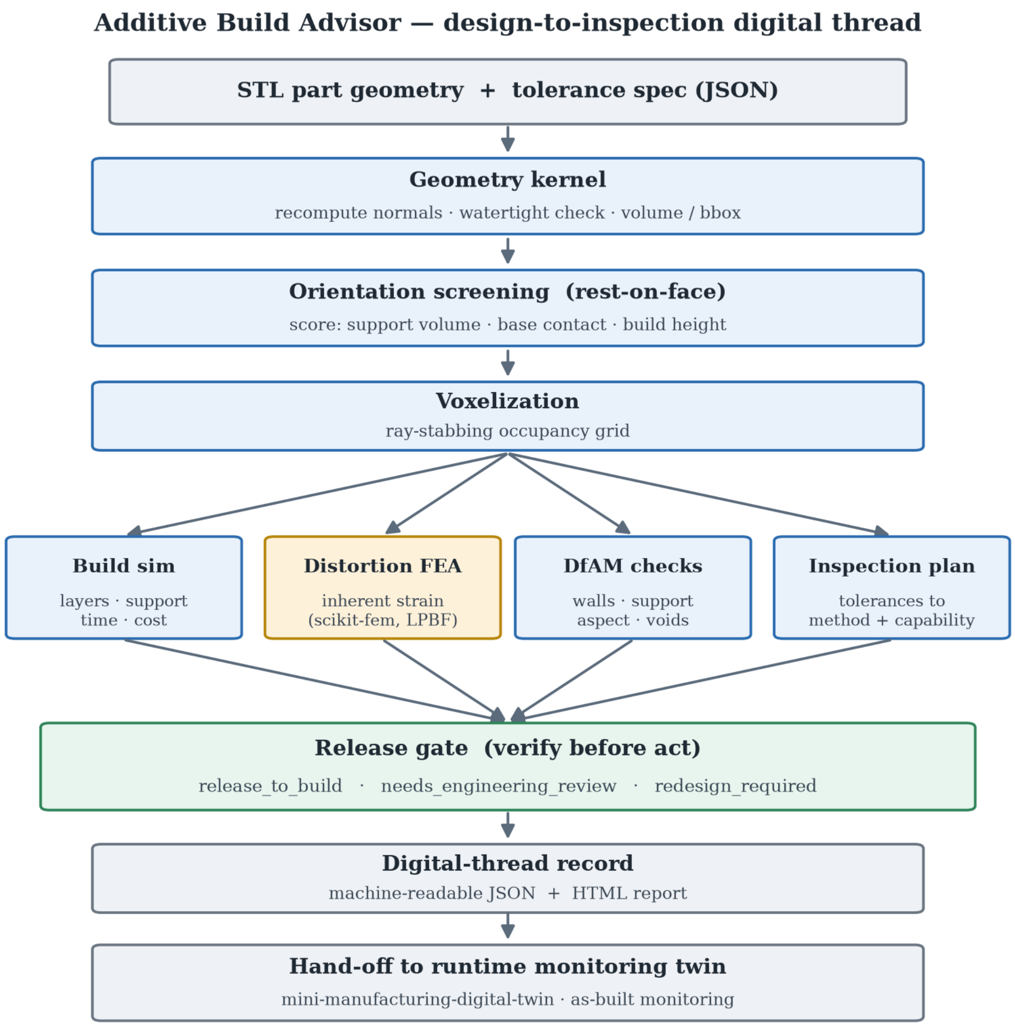

Figure 1: System diagram

1.5 Orientation: rest on a flat face

Orientation is the highest-leverage additive decision, so it deserves a physically meaningful search. Candidates are generated the way an engineer chooses — *which face goes down* — by clustering the mesh's facet normals into its significant flat faces and adding the six bounding-box directions as a fallback. Every candidate therefore rests a real face on the plate; there are no degenerate edge-balanced orientations.

Each candidate is screened on a coarse voxelization and scored on:

- **support material volume** (minimize) — the dominant cost/quality driver;
- **base-contact area** (maximize) — large flat footprint = adhesion + stability;
- **build height** (minimize) — drives build time;

with a hard penalty for any orientation that exceeds the build volume. On the sample bracket, the winner rests on the large flat back face: full base contact and **zero support**, versus alternatives that need >5 cm^3 of support on a tiny contact patch. The cheap coarse screen ranks the candidate set; the full simulation and FEA then run once, on the winner.

> An earlier version scored orientation only by overhang *facet area* and chose a 45° edge-balanced tilt — it drove the overhang metric to zero but was physically nonsense (no flat base, huge support). Scoring by real support volume and base contact fixed it. The lesson — a metric that is cheap to game is the wrong metric — is itself worth discussing.

1.6 Distortion FEA: the inherent-strain method

Distortion is predicted with a genuine finite-element solve, not a heuristic, using **scikit-fem** (an established FEM library) rather than a hand-rolled solver:

- the occupied voxels become a hexahedral mesh of 8-node trilinear (`ElementHex1`) vector elements — good for bending, unlike linear tets;
- the accumulated thermal shrinkage of the build is modeled as a uniform **eigenstrain** (inherent strain), entering as the consistent load `f = integral( sigma0 : sym_grad(v) )` with `sigma0 = (3*lam + 2*mu)*eps*`;
- the base layer is clamped to the build plate;
- the sparse system `K u = f` is assembled by scikit-fem and solved with SciPy's sparse direct solver; element von Mises stress is recovered for context.

The displacement field is the predicted distortion; its peak magnitude is the warpage estimate. Clamping the base while the bulk shrinks reproduces the corner-lift that dominates real additive distortion. This is the standard reduced-order approach used by Netfabb and ANSYS Additive.

A subtlety worth stating: for an eigenstrain-only load (no external force), the *displacement* field is independent of Young's modulus (it cancels between `K` and `f`), so distortion is governed by geometry, eigenstrain, and Poisson's ratio. Young's modulus only enters the stress field (peak von Mises is reported separately). That peak stress is linear-elastic and *indicative* only: with no plasticity in the model it can exceed the material yield, where a real part would yield and stress-relieve. The distortion field is the meaningful output; the stress is a relative flag.

The result is visualized as a **deformed element mesh** — the voxel surface displaced by the nodal solution (exaggerated for visibility) and contour-colored by displacement magnitude — rather than a node cloud, so the warpage is legible.

1.6.1 Target process, basis, and prior art

The distortion analysis targets **metal laser powder-bed fusion (LPBF)**, where residual-stress warpage is the governing failure mode. The inherent-strain method — calibrate an eigenstrain (originally from Ueda's inherent-strain theory, adapted to AM) and apply it as a static load to a part-scale elastic FEA — is the accepted part-scale approach and what commercial tools (Autodesk Netfabb, ANSYS Additive, Simufact) implement; see the state-of-the-art review in *Int. J. Adv. Manuf. Technol.* (2022). The recognized validation artifact is the **NIST AM-Bench 2018 (AMB2018-01)** single cantilever / 12-leg bridge, built in IN625 and 15-5 PH, measured by CMM before and after EDM release. The repo includes the cantilever geometry (`shapes.cantilever_benchmark`) and an IN625 profile to run on it. For polymer processes the same solver runs, but the record labels the result *indicative only*, since the eigenstrain calibration is a metal-PBF notion.

1.6.2 What the number means: on-plate vs post-release

An important, honestly-stated limitation. This model holds the part **bonded to the build plate** (base clamped) and reports the resulting *on-plate* distortion field. The NIST cantilever's headline ~1.0–1.3 mm is the **post-release** deflection measured *after* the part is cut from the plate, when stored residual stress relaxes into a large curl — a different and larger quantity. Our predicted on-plate cantilever distortion (~0.1 mm) is therefore not directly comparable to that 1.3 mm, and the project says so rather than fudging the match. Capturing the post-release deflection requires a build-plate release/cutting step and a calibrated (anisotropic, layer-activated) inherent strain — listed under "Production extension plan." What *is* validated rigorously is the solver itself, below.

1.6.3 FEA validation

Analytical clamped bar. A prismatic bar clamped at the base under uniform eigenstrain ε^* has an analytical top displacement $|\varepsilon^*| \cdot \mathrm{H}$. Refining the mesh, the FEA converges toward it:

voxel pitch	FEA peak distortion	error
4.0 mm	0.4213 mm	+5.3%
2.0 mm	0.4161 mm	+4.0%
1.0 mm	0.4141 mm	+3.5%
0.5 mm	0.4132 mm	+3.3%

(analytic axial value 0.400 mm for $\varepsilon^* = -0.01$, H = 40 mm). The residual few percent is the lateral corner motion — the reported metric is the peak displacement *magnitude*, which at the top corner includes the in-plane shrinkage on top of the axial component; the axial component itself converges to 0.400 mm.

Eigenstrain linearity / E-independence. Running the same bracket across every process, predicted distortion scales linearly with the inherent strain and is independent of Young's modulus — ABS ($\varepsilon^* = -0.012$) gives 0.602 mm, exactly twice PLA's 0.302 mm ($\varepsilon^* = -0.006$), and metal Ti-6Al-4V (E = 110 GPa) and polymer PLA (E = 3.5 GPa) sit purely where their eigenstrains place them. That linear, E-independent response is exactly what linear elasticity requires for an eigenstrain-only load, and is the second validation that the solver is correct.

1.7 Build simulation

From the oriented mesh and grid, the simulation estimates layer count (from the real layer height), support material (support infill × the empty volume sitting under solid), build time (deposition + per-layer recoat/peel overhead), and cost (material + amortized machine time). These scale correctly across processes: the same bracket is ~160 layers and a few dollars on FFF but ~1000 layers and a few hundred dollars on metal LPBF, driven by the 0.03 mm layer height and machine rate.

1.8 DfAM checks

Transparent, severity-ranked manufacturability rules, all reading from the same grid, simulation, and FEA as the headline numbers:

Check	Method
Build-volume fit	bounding box vs machine envelope
Thin walls	morphological opening at the min-wall radius
Support burden	support material as a fraction of part volume
Aspect ratio	height / min footprint dimension
Trapped volume	flood fill from outside; unreachable voids are enclosed
Distortion	FEA peak displacement as a fraction of the part's largest dimension

1.9 Inspection planning and process capability

The planner turns each toleranced dimension, GD&T control, and surface-finish requirement into an inspection step with a method and equipment chosen by how tight the tolerance is (calipers → micrometer → CMM → CT), and checks each tolerance against the process's **as-built capability** — flagging any tolerance that needs post-machining, because inspecting to an impossible tolerance only confirms a guaranteed failure. The tolerance spec is plain JSON, so the design data stays CAD-neutral (a Fusion or STEP exporter would populate the same fields).

1.10 The release gate (verify before act)

The advisor never silently approves a build. It returns one of three decisions with reasons and blocking findings attached:

Decision	When
`release_to_build`	DfAM clean, tolerances within capability, simulation validated
`needs_engineering_review`	warnings, tolerances needing post-machining, or low simulation confidence (unsealed mesh / coarse-grid volume error)
`redesign_required`	a critical DfAM finding (does not fit, traps resin/powder, severe thin walls)

This is the same verify-before-act discipline used in the companion runtime monitoring twin and in prior robotic-manipulation work: a model may recommend, but a physical action is gated on evidence, on the confidence of the model, and on a human-review path when anything is uncertain. The record carries an explicit hand-off block so design intent flows into as-built monitoring.

Worked outcomes from the sample run: **calibration_cube / FFF** → `release_to_build`; **gantry_bracket / FFF** → `needs_engineering_review` (±0.05 mm and 3.2 µm are below FFF capability → route to machining); **hollow_housing / SLA** → `redesign_required` (enclosed cavity traps resin → add drain holes).

1.11 Validation summary

```
pytest                        # 10 tests
python tests/test_smoke.py    # -> "Smoke test passed"
python examples/validate_fea.py    # regenerates the FEA validation figure
```

The tests cover: sample parts are watertight; STL round-trips; the cube discretizes exactly; an off-axis part's volume converges; trapped volume is detected; the chosen orientation rests on a real face and is not the highest-support candidate; the FEA matches the analytical clamped-bar solution within 10%; distortion appears in the record; all three gate outcomes fire; and the record is JSON-clean.

1.12 Limitations

This prototype is intentionally scoped. It does **not** include:

- A true slicer / toolpath generator (build time is volume- and layer-based).
- A calibrated transient thermo-mechanical solve. The FEA is a real linear- elastic solve, but the inherent strain is a representative per-process value, not fit to melt-pool thermal history.
- OEM-qualified material/machine profiles (numbers are representative defaults).
- A real CAD/CAM integration (geometry is STL; tolerances are JSON).
- Lattice/infill modeling, multi-part nesting, or layer-by-layer activation in the FEA (the eigenstrain is applied to the whole part at once).
- Validation against measured build outcomes.

These limits are stated plainly because the value here is the architecture and the engineering judgment, not a claim of production fidelity.

1.13 Production extension plan

1. A real slicer with toolpath-length-based time and proper support generation.
2. Layer-by-layer element activation and a calibrated inherent-strain (or transient thermo-mechanical) solve, validated against measured distortion.
3. Qualified per-machine process profiles and measured-vs-predicted calibration.
4. CAD/CAM integration — Fusion 360 / Autodesk Platform Services or STEP — to pull features, datums, and PMI directly into the same record schema.
5. Closed-loop feedback: as-built inspection and in-situ monitoring flowing back to update capability data and the eigenstrain calibration.
6. A persisted thread store linking design revision → build → inspection → part.

1.14 Interview explanation

"I built this for the front half of a digital thread — design intent turning into a build decision — grounded in real additive physics. It takes an STL, picks a build orientation by resting the part on its flat faces and scoring real support volume and base contact, simulates the build on a voxel model I validate against analytic volume, and predicts distortion with an actual finite-element solve: a linear-elastic voxel FEM using the inherent-strain method, clamped at the plate, solved with scikit-fem. I validated the FEA against the analytical clamped-bar case and confirmed the distortion is linear in eigenstrain and E-independent, which is what the theory demands. Then DfAM and inspection- capability checks feed a gate that decides release, review, or redesign. I kept the eigenstrain a representative value rather than pretending it's a calibrated transient thermo-mechanical solve — I've run the production FEA and I'm precise about what this is and isn't."

1.15 Strong follow-up line

"The orientation step is a good example of how I work. My first version scored it by overhang facet area and it picked a 45° edge-balanced tilt — mathematically zero overhang, physically absurd. The fix was to score the thing that actually costs money: support volume and base contact from a real voxelization. A metric that's cheap to game is the wrong metric, and catching that is exactly the judgment this kind of applied research needs."